

Submission A solution to Problem 8

A polar transform for the relative Dressian

0. Conventions and two local facts

Let E be the ground set, $r = \text{rk } M$, and μ_M the trivial rank r valuated matroid supported on the bases of M . I use the standard model of the relative Dressian: a rank k point is a projective valuated matroid ν with $\nu \preceq \mu_M$, where $\preceq$ denotes valuated quotient. Brandt–Eur–Zhang [1] identify this order with inclusion of projective tropical linear spaces,

$$\text{Trop}(\theta) \subseteq \text{Trop}(\nu) \iff \theta \preceq \nu.$$

For valuated matroids p, q of ranks $a \leq b$, $p \preceq q$ is equivalent to the incidence-Plucker relations

$$\min_{i \in J \setminus I} \{p(I \cup i) + q(J \setminus i)\} \text{ attained at least twice} \quad (\text{IP})$$

for all $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$. All- ∞ minima are regarded as attained at every displayed index.

Criterion 1 (nested incidence suffices). *Let p, q be valuated matroids of ranks $a < b$ whose ordinary supports are in quotient order. If (IP) holds whenever $I \subseteq J$, then (IP) holds for all I, J .*

Proof. For a rank a valuated matroid p and an $(a-1)$ -set I , write $C_p^*(I)_e = p(I \cup e)$, with value ∞ if $e \in I$. We use the standard cocircuit-generation theorem of Dress–Wenzel/Speyer [2, 5]: the tropical linear space of a valuated matroid is the closed tropical convex hull of its valuated cocircuits.

Fix arbitrary $I \in \binom{E}{a-1}$ and $J \in \binom{E}{b+1}$, and put $c_j = q(J \setminus j)$ for $j \in J$. If all c_j are ∞ there is nothing to prove. Otherwise J contains a q -basis. Since the ordinary support of p is a quotient of the ordinary support of q , every q -basis spans the support matroid of p (indeed, $a - \text{rk}_p(Q) \leq b - \text{rk}_q(Q) = 0$ for a q -basis Q); hence the restriction $p|_J$ has rank a .

For every $S \subseteq J$ with $|S| = a-1$, the nested hypothesis with upper set J says precisely that the cocircuit $C_{p|_J}^*(S)$ is tropically orthogonal to $c = (c_j)_{j \in J}$. The set of vectors orthogonal to a fixed c is a tropical hyperplane, hence is tropically convex; therefore it contains the

whole tropical linear space of $p|J$. It remains only to note that $C_p^*(I)|_J$ is a vector of $p|J$: for each valuated circuit of $p|J$, indexed by a set $D \in \binom{J}{a+1}$, the orthogonality relation

$$\min_{e \in D} \{p(I \cup e) + p(D \setminus e)\}$$

is exactly a Plucker relation for p . Hence $C_p^*(I)|_J$ is orthogonal to c , which is the desired arbitrary incidence relation. $\square$

We also use Wang's codimension-one theorem in the following precise form. In Wang's terminology the "three-term incidence relations" consist of degenerate, parallel, and concurrent hyperplane relations; the displayed tropical minimum below is only the concurrent relation.

Proposition 1 (elementary quotient test). *Let η be a rank d valuated matroid and let $\theta : \binom{E}{d-1} \rightarrow \mathbb{R} \cup \{\infty\}$ be nonzero. Suppose that*

$$\theta(A) = \infty \quad \text{whenever } A \text{ is dependent in the support matroid of } \eta \quad (\text{D})$$

and, when $d \geq 2$, for every $S \in \binom{E}{d-2}$ and distinct $x, y, z \notin S$,

$$\min\{\theta(Sx) + \eta(Syz), \theta(Sy) + \eta(Sxz), \theta(Sz) + \eta(Sxy)\} \quad (\text{C})$$

is attained at least twice. Then θ is a valuated matroid and $\theta \preceq \eta$.

Proof. For $d = 1$ this is the unique rank-zero quotient. Let $d \geq 2$. By Wang's Theorem 4.3 [6], $\text{Dr}^1(\eta)$ is cut out by the degenerate, parallel, and concurrent three-term incidence relations. Condition (D) is exactly the degenerate part, and (C) contains the concurrent part. It remains only to see that the parallel part follows from (C). Write $A = Di$ and $B = Dj$ for independent $(d-1)$ -sets in the support matroid of η with the same closure, and choose k outside that closure, so Dik and Djk are bases. Applying (C) to D, i, j, k , the third upper coordinate $\eta(Dij)$ is ∞ , while $\eta(Dik)$ and $\eta(Djk)$ are finite. Hence

$$\theta(Di) + \eta(Djk) = \theta(Dj) + \eta(Dik),$$

which is Wang's parallel hyperplane relation. Thus all of Wang's defining relations hold, so θ represents a codimension-one tropical linear subspace of $\text{Trop}(\eta)$. $\square$

Finally, if $\nu \preceq \mu_M$ has rank k , we use its valuated Higgs completion

$$h_j(B) = \begin{cases} \min\{\nu(C) : B \subseteq C, |C| = k\}, & j \leq k, |B| = j, \\ \min\{\nu(C) : C \subseteq B, |C| = k\}, & j \geq k, B \in \mathcal{I}_j(M), \\ \infty, & j \geq k, B \notin \mathcal{I}_j(M). \end{cases} \quad (1)$$

Murota's truncation and relative elongation theorems [4, Theorems 2.1 and 4.1], equivalently the Dress–Terhalle well-layering construction [3], give valuated matroids h_j and adjacent quotients $h_j \preceq h_{j+1}$. The top layer is projectively μ_M : if $B' = B - e + f$ are adjacent M -bases and $C \subseteq B$ minimizes (1), then either $e \notin C$ and $C \subseteq B'$, or the incidence relation for $\nu \preceq \mu_M$ with lower set $C - e$ and upper set $B \cup f$ gives a k -subset of B' of value at most $\nu(C)$; by symmetry the two top-layer values are equal.

1. Flat-lattice duality and flat constancy

Lemma 1. *If $\mathcal{L}(M)$ admits an order-reversing involution $X \mapsto X^\perp$, then $\mathcal{L}(M)$ is modular and $\text{rk}(X^\perp) = r - \text{rk}(X)$.*

Proof. The anti-automorphism reverses maximal chains, so it sends rank i to rank $r - i$. Applying it to the semimodular inequality for the geometric lattice $\mathcal{L}(M)$ gives the reverse inequality, hence equality. $\square$

Lemma 2 (flat constancy). *Let $\nu \preceq \mu_M$ have rank k . If B, B' are independent k -subsets of M with $\text{cl}(B) = \text{cl}(B')$, then $\nu(B) = \nu(B')$; and $\nu(A) = \infty$ whenever A is dependent in M .*

Proof. The support matroid of ν is an ordinary quotient of M , so its bases are independent in M . For $k > 0$ it suffices to compare $B = A \cup b$ and $B' = A \cup b'$ spanning the same rank k flat. Choose one complement C so that both AbC and $Ab'C$ are M -bases. In the incidence relation for $\nu \preceq \mu_M$ with lower set A and upper set $A \cup \{b, b'\} \cup C$, the only finite μ_M -terms are the b - and b' -terms. Thus $\nu(Ab) = \nu(Ab')$. $\square$

Write $\bar{\nu}(X)$ for the common value on independent k -sets spanning the rank k flat X , and put $\bar{\nu}(X) = \infty$ if no basis of ν spans X .

2. The polar transform

For a rank k quotient $\nu \preceq \mu_M$ define

$$\nu^{\perp M}(A) = \begin{cases} \bar{\nu}((\text{cl } A)^\perp), & A \text{ independent in } M, |A| = r - k, \\ \infty, & \text{otherwise.} \end{cases} \quad (2)$$

This is projectively well-defined, and applying it twice recovers the original flat coordinates.

Lemma 3 (ordinary polar rank). *Let N be an ordinary quotient of M of rank k , with rank function ρ_N . Then*

$$\rho^\perp(A) = \text{rk}_M(A) - k + \rho_N((\text{cl } A)^\perp) \quad (3)$$

is a matroid rank function. Its bases are precisely the finite support of (2). If N_p is a quotient of N_q , then $N_q^{\perp M}$ is a quotient of $N_p^{\perp M}$.

Proof. Since N is a quotient of M , ρ_N is constant on M -closures: if $e \in \text{cl}_M(A)$, then the N -rank increment from A to Ae is at most the corresponding M -increment, namely 0. First restrict (3) to flats. For flats $X \leq Y$,

$$\rho^\perp(Y) - \rho^\perp(X) = \text{rk}(Y) - \text{rk}(X) - (\rho_N(X^\perp) - \rho_N(Y^\perp))$$

is between 0 and $\text{rk}(Y) - \text{rk}(X)$ by the strong-map increment inequality, so $\rho^\perp$ is monotone with unit increments on flats. For flats X, Y , modularity of rk_M and the identities $(X \wedge Y)^\perp = X^\perp \vee Y^\perp$ and $(X \vee Y)^\perp = X^\perp \wedge Y^\perp$ reduce submodularity of $\rho^\perp$ to submodularity of ρ_N .

For arbitrary subsets, $\rho^\perp(A) = \rho^\perp(\text{cl}_M A)$. If $X = \text{cl } A$ and $Y = \text{cl } B$, then $\text{cl}(A \cap B) \leq X \wedge Y$ and $\text{cl}(A \cup B) = X \vee Y$; by monotonicity on flats, submodularity for X, Y implies submodularity for A, B . The unit-increment property also passes from flats to subsets. Thus (3) is a matroid rank function. An M -independent set A of size $r - k$ has full $\rho^\perp$ -rank exactly when $\rho_N((\text{cl } A)^\perp) = k$, and M -dependent sets cannot be bases; this is precisely the finite support of (2).

If N_p is a quotient of N_q , then on every interval the rank increment of N_p is at most that of N_q . The displayed formula for increments therefore reverses the inequality for the polar ranks, which is exactly the strong-map criterion saying that $N_q^{\perp M}$ is a quotient of $N_p^{\perp M}$. $\square$

Lemma 4 (elementary opposition). *Let $p \preceq q \preceq \mu_M$, with $\text{rk } p = d$ and $\text{rk } q = d + 1$. Then $q^{\perp M}$ and $p^{\perp M}$ satisfy all adjacent three-term relations of ranks $r - d - 1$ and $r - d$.*

Proof. Fix A of size $r - d - 2$ and distinct $x, y, z \notin A$. Put $U = \text{cl}(A)$, $W = \text{cl}(Axyz)$, and $\delta = \text{rk } W - \text{rk } U$. If A is dependent, or if $\delta \leq 1$, each term contains a dependent set of the required size and hence is ∞ .

If $\delta = 2$, let $G = W^\perp$ and $H = U^\perp$, of ranks d and $d + 2$. Whenever the term indexed by x is finite, Ayz spans W , so its p -factor is the common value $\bar{p}(G)$; its q -factor is $\bar{q}((\text{cl}(Ax))^\perp)$, and similarly for y, z . In the rank-two interval $[U, W]$, a loop over U makes all terms involving it infinite, and two parallel elements give equal corresponding q -coordinates. Thus only the case of three non-parallel points over U needs proof. Choose a basis S of G , representatives a_x, a_y, a_z for the three dual rank-one flats over G in $[G, H]$, and a complement C of H to a basis of M . In the incidence relation for $q \preceq \mu_M$ with lower set S and upper set $S \cup \{a_x, a_y, a_z\} \cup C$, the only finite μ_M -terms are those obtained by deleting one of a_x, a_y, a_z . Hence the three displayed q -coordinates have a non-unique minimum; adding the common $\bar{p}(G)$ proves the transformed relation.

If $\delta = 3$, then x, y, z form a Boolean frame over U . Let $G = W^\perp$ and

$$P_x = (\text{cl}(Ayz))^\perp, \quad P_y = (\text{cl}(Axz))^\perp, \quad P_z = (\text{cl}(Axy))^\perp.$$

These are the three atoms over G in the dual frame, and $(\text{cl}(Ax))^\perp = P_y \vee P_z$, with the analogous identities for y, z . Choose a basis S of G and representatives $a_e \in P_e \setminus G$. The adjacent incidence relation for $p \preceq q$ applied to the lower set S and to $S \cup \{a_x, a_y, a_z\}$ is exactly the transformed three-term relation, using flat constancy to identify the displayed coordinates. $\square$

Proposition 2. *The formula $\nu \mapsto \nu^{\perp M}$ defines an involution of the set underlying $\text{Dr}(M)$.*

Proof. Insert ν in the complete flag (1). We prove inductively that

$$h_r^{\perp M} \preceq h_{r-1}^{\perp M} \preceq \cdots \preceq h_0^{\perp M} = \mu_M.$$

After harmless scalar normalization, the right endpoint $h_0^{\perp M} = \mu_M$ is clear from (2). If $r \leq 1$, there is nothing more to check beyond the automatic rank-zero quotient. Now assume $r \geq 2$. Suppose $0 \leq j \leq r - 2$ and $h_j^{\perp M}$ has already been proved valuated. Since $h_j \preceq h_{j+1}$, Lemma

3 says that the ordinary support of $h_{j+1}^{\perp M}$ is an elementary quotient of the ordinary support of $h_j^{\perp M}$; in particular it satisfies the degenerate support condition (D) relative to $h_j^{\perp M}$. Lemma 4 gives the displayed concurrent relations (C) for the pair $h_{j+1}^{\perp M}, h_j^{\perp M}$. Proposition 1 then implies $h_{j+1}^{\perp M} \preceq h_j^{\perp M}$. This proves the induction down to $h_{r-1}^{\perp M}$; the last rank-zero quotient $h_r^{\perp M} \preceq h_{r-1}^{\perp M}$ is automatic. Hence $\nu^{\perp M}$, one layer of the flag, lies in $\text{Dr}(M)$. Involutivity follows from (2) and $(X^\perp)^\perp = X$. $\square$

3. Order reversal

Lemma 5 (circuit minima). *Let $w \preceq \mu_M$ have rank $k \geq 1$, let A be an independent $(k-1)$ -set of M , and let C be a circuit of M/A with $|C| \geq 2$. Then $\min_{i \in C} w(A \cup i)$ is attained at least twice.*

Proof. Let $F = \text{cl}(A \cup C)$. For every $i \in C$, $A \cup (C - i)$ is independent and spans F . Choose one complement D of F to an M -basis. In the incidence relation for $w \preceq \mu_M$ with lower set A and upper set $A \cup C \cup D$, the finite μ_M -terms are exactly those indexed by $i \in C$. $\square$

Theorem 1. *If $p \preceq q \preceq \mu_M$, then*

$$q^{\perp M} \preceq p^{\perp M}.$$

Proof. Let $\text{rk } p = a$ and $\text{rk } q = b$. If $a = b$, the strong-map inequalities applied to $\emptyset \subseteq A$ and to $A \subseteq E$ force the two ordinary support rank functions to be equal. Then p and q are projectively equal: the equal-rank incidence relation on adjacent bases shows $p(B) - q(B)$ is constant on the common basis graph. If $a = 0$, then $p^{\perp M} = \mu_M$, so the assertion is $q^{\perp M} \preceq \mu_M$; if $b = r$, it is the vacuous rank-zero lower quotient. Hence assume $0 < a < b < r$.

By Proposition 2 the polars are valuated matroids, and by Lemma 3 their ordinary supports are in quotient order. By Criterion 1 it remains to prove nested incidence. Fix $A \subseteq T$ with $|A| = r - b - 1$, $|T| = r - a + 1$, set $B = T \setminus A$, and $m = |B| = b - a + 2$. If A is dependent all terms are ∞ . Otherwise put $U = \text{cl}(A)$, $W = \text{cl}(T)$, and $\delta = \text{rk } W - \text{rk } U$. A finite term forces $T - i$ to be independent, hence $\delta \geq m - 1$; since $\delta \leq m$, only two cases remain.

If $\delta = m$, then T is independent over U . Let $G = W^\perp$ and $H = U^\perp$, of ranks $a - 1$ and $b + 1$. The interval generated by B is Boolean; after applying $^\perp$, the flats $X_i = (\text{cl}(T - i))^\perp$ are the atoms over G , and $Y_i = (\text{cl}(Ai))^\perp$ is the join of all X_j with $j \neq i$. Choose a basis S of G and representatives $x_i \in X_i \setminus G$. The nested incidence relation for $p \preceq q$ with lower set S and upper set $S \cup \{x_i : i \in B\}$ is exactly the desired transformed relation.

If $\delta = m - 1$, then B has nullity one in M/A , hence a unique circuit C . A loop circuit gives only ∞ terms. Otherwise, for $i \notin C$ the set $T - i$ is dependent; for $i \in C$ it is independent and spans W , so the $p^{\perp M}(T - i)$ factor is the common value $\bar{p}(W^\perp)$. Lemma 5, applied to $w = q^{\perp M}$ and the circuit C of M/A , gives two minima among the remaining $q^{\perp M}(Ai)$ terms. This proves nested incidence. $\square$

4. Agreement with flats

Let F be a flat of rank f , and let q_F be the trivial rank $r - f$ valuated matroid of $M/F \oplus U_{0,F}$. For an independent $(r - f)$ -set B with $Z = \text{cl}(B)$,

$$q_F(B) < \infty \iff Z \vee F = \hat{1}.$$

For an independent f -set A with $Y = \text{cl}(A)$, formula (2) gives

$$q_F^{\perp M}(A) < \infty \iff Y^{\perp} \vee F = \hat{1}.$$

By modularity, since $\text{rk}(Y^{\perp}) = r - f$ and $\text{rk} F = f$, this is equivalent to $Y^{\perp} \wedge F = \hat{0}$. Applying the anti-automorphism gives $Y \vee F^{\perp} = \hat{1}$, the basis condition for $M/F^{\perp} \oplus U_{0,F^{\perp}}$. The finite values are zero on both sides. Thus the polar transform sends the flat F to $F^{\perp}$.

Combining Proposition 2, Theorem 1, and this flat computation, the map

$$\Phi([\nu]) = [\nu^{\perp M}]$$

is an order-reversing involution of $\text{Dr}(M)$ extending the given order-reversing involution of $\mathcal{L}(M)$.

References

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